UE24MA241B LINEAR ALGEBRA AND ITS APPLICATIONS

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LINEAR ALGEBRA AND ITS APPLICATIONS

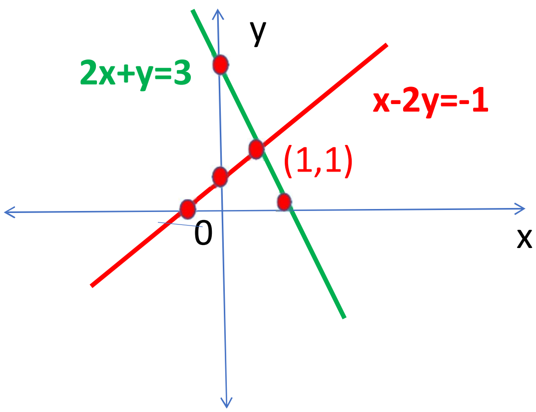
MATRICES AND GAUSSIAN ELIMINATION

slides by Renna Sultana, Department of Science and Humanities

Course Content: The Geometry of Linear Equations

1. Solve the system of equations 2x+y=3; x-2y=-1 and draw the Row picture and Column picture.

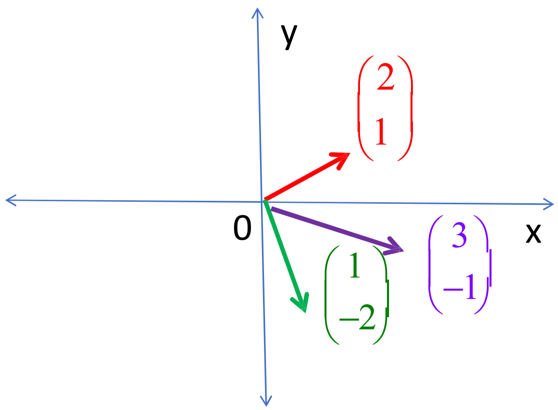
Row picture



Intersecting lines
Unique solution (1,1)

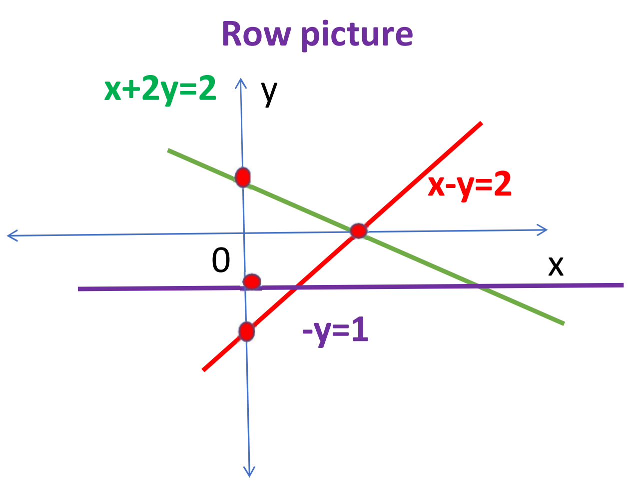
Column picture

$$x\begin{pmatrix}2\\1\end{pmatrix}+y\begin{pmatrix}1\\-2\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}\Rightarrow 1\begin{pmatrix}2\\1\end{pmatrix}+1\begin{pmatrix}1\\-2\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$$



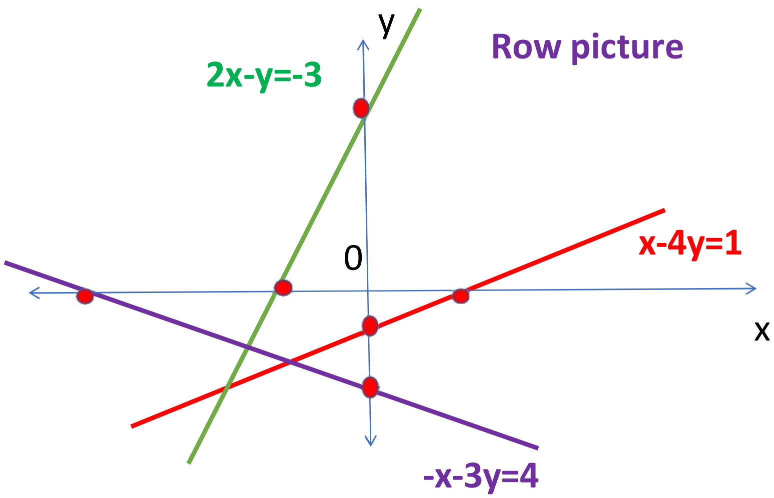
THE GEOMETRY OF LINEAR EQUATIONS:

2. Sketch the three lines x+2y=2; x-y=2; -y=1(row picture only) and decide if the three equations are solvable. What happens if all right hand sides are zero? Is there any non-zero choice of right hand sides that allow the three lines to intersect at a common point of intersection?



The given three equations are not solvable as they do not have a common point of intersection. If all right hand sides are zero we get a trivial(zero) solution i.e x=0, y=0. If RHS of first equation is -1 then the three lines x+2y=-1; x-y=2; -y=1 intersect at a common point of intersection (1,-1)

3. Sketch the three lines $x - 4y = 1$; $2x - y = -3$; $-x - 3y = 4$ and decide if the three equations are solvable. Do they have a common point of intersection? Explain.

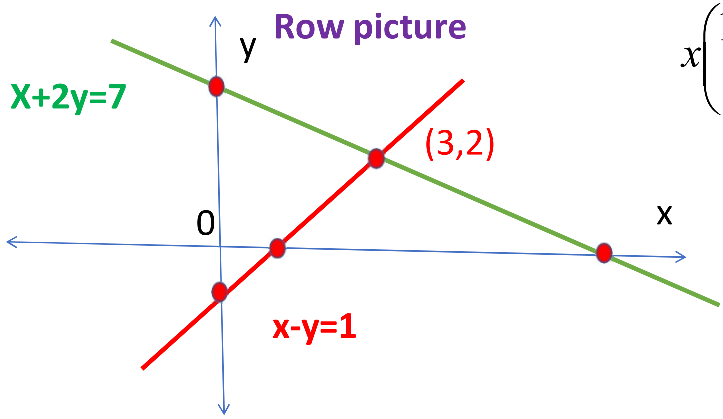


Given three equations are not solvable as they have no common point of intersection.

4. Draw the row picture and column picture for the following system of equations and discuss its consistency, singularity, and existence of the solution:

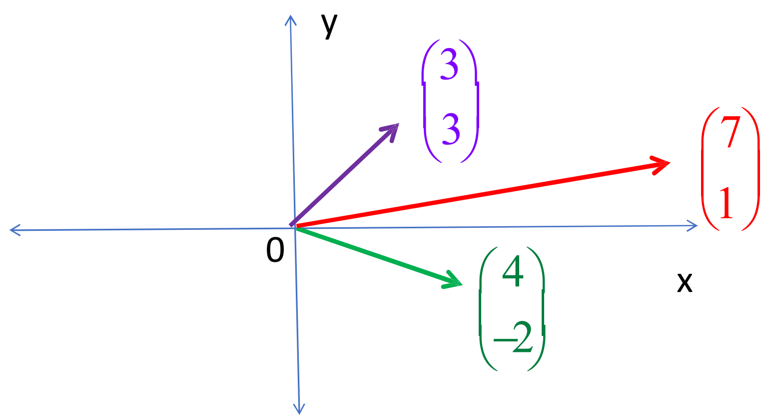
(i) x + 2y = 7; x - y = 1

Row picture



Column picture

$$x\begin{pmatrix}1\\1\end{pmatrix}+y\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}7\\1\end{pmatrix}\Rightarrow 3\begin{pmatrix}1\\1\end{pmatrix}+2\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}7\\1\end{pmatrix}$$



(ii) 2x + 3y = 6; 4x + 6y = 12 (iii) x - 3y = 5; x - 3y = -5

INTRODUCTION:

Example:

$$A \rightarrow \begin{pmatrix} 1 & 1 & -2 & 3: & 4 \\ 2 & 3 & 3 & -1: & 3 \\ 5 & 7 & 4 & -1: & 5 \end{pmatrix} \xrightarrow[R_3-5R_1]{R_2-2R_1} \begin{pmatrix} 1 & 1 & -2 & 3: & 4 \\ 0 & 1 & 7 & -7: & -5 \\ 0 & 2 & 14 & -16: & -15 \end{pmatrix} \xrightarrow{R_3-2R_2}$$

$$\begin{pmatrix} 1 & 1 & -2 & 3: & 4 \\ 0 & 1 & 7 & -7: & -5 \\ 0 & 0 & 0 & -2: & -5 \end{pmatrix} = U \xrightarrow{R_3/(-2)} \begin{pmatrix} 1 & 1 & -2 & 3: & 4 \\ 0 & 1 & 7 & -7: & -5 \\ 0 & 0 & 0 & 1: & 5/2 \end{pmatrix} \xrightarrow[R_2+7R_3]{R_1-3R_3}$$

$$\begin{pmatrix} 1 & 1 & -2 & 0: & -7/2 \\ 0 & 1 & 7 & 0: & 25/2 \\ 0 & 0 & 0 & 1: & 5/2 \end{pmatrix} \xrightarrow{R_1-R_2} \begin{pmatrix} 1 & 0 & -9 & 0: & -16 \\ 0 & 1 & 7 & 0: & 25/2 \\ 0 & 0 & 0 & 1: & 5/2 \end{pmatrix} = R$$

ENGINEERING MATHEMATICS-III

References/Links:

https://upload.wikimedia.org/wikipedia/commons/c/c0/Intersecting_Lines.svg

Google search: Graphs of row and column pictiure for a system of linear equations

THANK YOU

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